

Expt 4

Design of Low Pass Filter using Windowing Techniques

4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

Objective:

1. To design an FIR Low-Pass Filter using the Rectangular Window technique for the given cutoff frequency, sampling frequency, and filter order specifications.
2. To generate the FIR filter coefficients using the Rectangular Window method in MATLAB and implement the designed filter.
3. To obtain and analyze the magnitude response and phase response of the FIR Low-Pass Filter designed using the Rectangular Window.

Procedure:

1. Open MATLAB and create a new script file.
2. Initialize the workspace using `clc`, `clear`, and `close all`.
3. Specify the filter design parameters: sampling frequency, cutoff frequency, and filter order.
4. Calculate the normalized cutoff frequency.
5. Design the FIR Low-Pass Filter using the `fir1()` function with a Rectangular Window (`rectwin`).
6. Generate and display the FIR filter coefficients.
7. Obtain the magnitude and phase responses using the `freqz()` function.
8. Plot and analyze the impulse response of the designed filter.
9. Generate the pole-zero plot and examine the filter characteristics.
10. Verify the stability and frequency-selective performance of the FIR Low-Pass Filter.

Program(Matlab Code):

```
clc;
```

```
clear;
```

```
close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000;          % Passband cutoff frequency (Hz)
```

```

fs = 4000;          % Sampling frequency (Hz)
N = 50;            % Filter order

wc = fp/(fs/2);    % Normalized cutoff frequency

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');
disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');
disp(b_hann);

%% --- Plot Frequency Responses ---

figure;

freqz(b_rect, 1);

title('Low-pass FIR using Rectangular Window');
grid on;

```

figure;

```
freqz(b_hamm, 1);
```

```
title('Low-pass FIR using Hamming Window');
```

```
grid on;
```

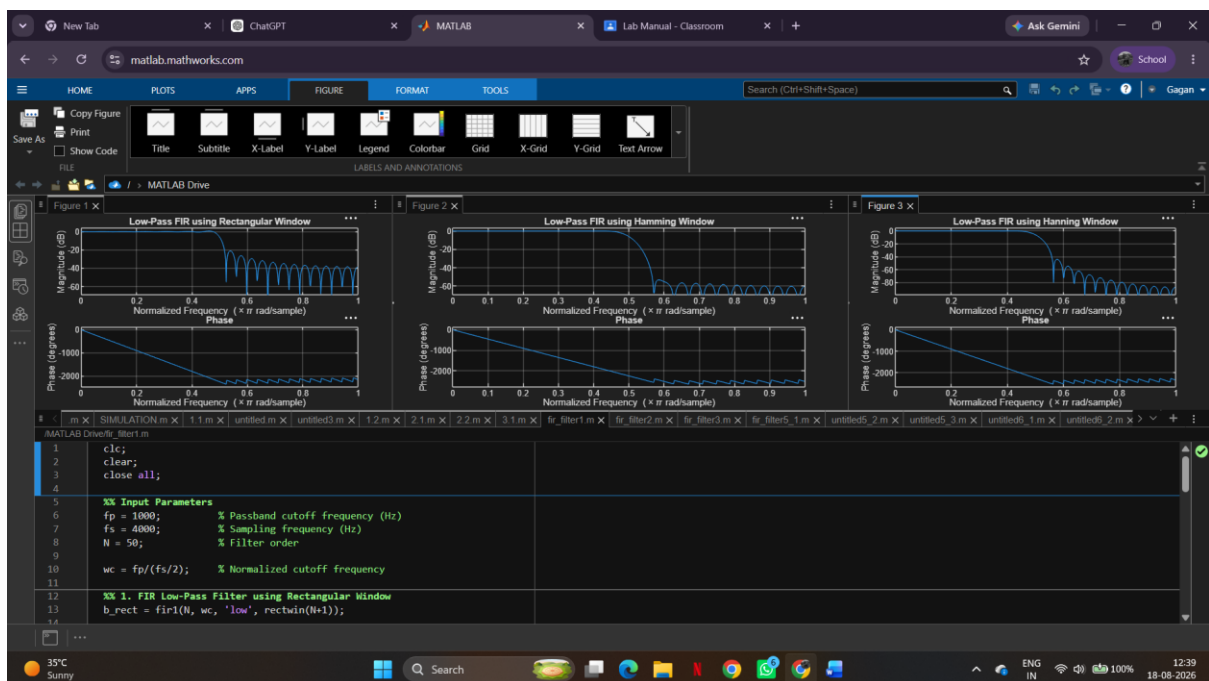
figure;

```
freqz(b_hann, 1);
```

```
title('Low-pass FIR using Hanning Window');
```

```
grid on;
```

OUTPUT:



--- Rectangular Window Coefficients ---

0.0126

-0.0000

-0.0137

0.0000

0.0150

-0.0000

-0.0166

0.0000

0.0185

-0.0000

-0.0210

0.0000

0.0242

-0.0000

-0.0286

0.0000

0.0349

-0.0000

-0.0449

0.0000

0.0629

-0.0000

-0.1048

0.0000

0.3145

0.4940

0.3145

0.0000

-0.1048

-0.0000

0.0629

0.0000

-0.0449

-0.0000

0.0349

0.0000

-0.0286

-0.0000

0.0242

0.0000

-0.0210

-0.0000

0.0185

0.0000

-0.0166

-0.0000

0.0150

0.0000

-0.0137

-0.0000

0.0126

Result:

1. An FIR Low-Pass Filter was successfully designed using the Rectangular Window technique for the specified cutoff frequency, sampling frequency, and filter order.
2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.
3. The magnitude response and phase response of the designed FIR Low-Pass Filter were obtained and analyzed. The magnitude response exhibited low-pass characteristics, while the phase response demonstrated the expected linear-phase behavior.